

DSC Calibration Handbook

What each parameter in the M3 stability-control firmware actually changes — by subsystem, with the algorithm it feeds and the direction of its effect.

Image 7846816A · 1 MiB **CPU** Freescale M·CORE (BE) **File offset** = CPU + 0x8000

Frame 10 ms **Maps to** xdf/MK60E5_7846816A.xdf

Scope. Static analysis of the owner's own ECU, for understanding and comparison. Stock values are quoted so edits are legible against a baseline. Not a recipe for disabling the safety system on a road car: the strongest levers (DSC mode, the active variant) are EEPROM, not flash, and every edited image must be re-signed to load.

Reading a parameter

Each table gives the parameter, its address, the stock value, and **what happens if you raise it**. Two tags carry the meaning:

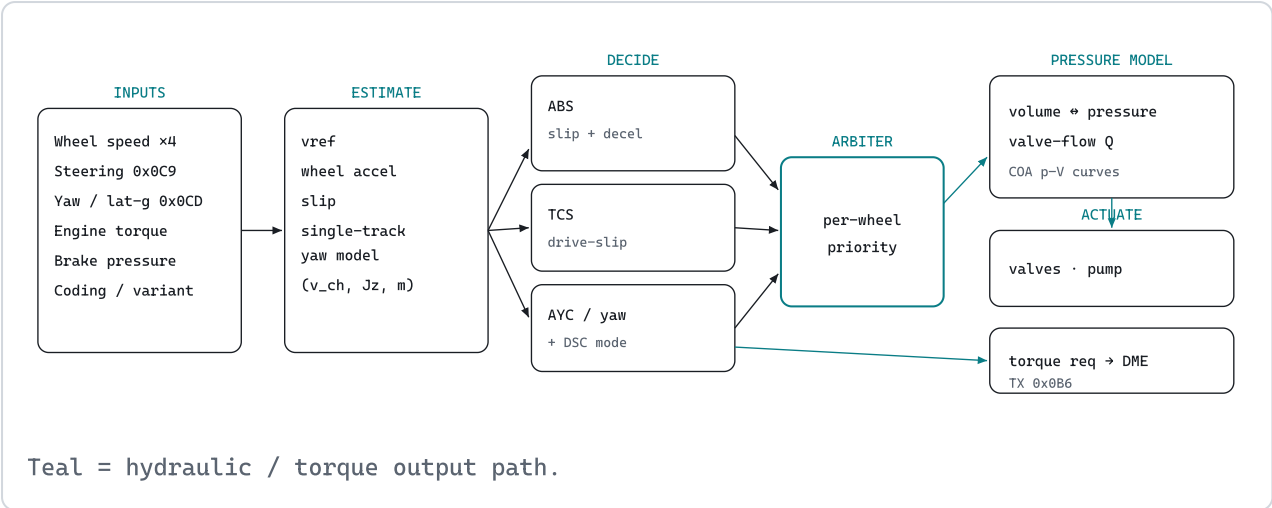
CAL in a calibration block — editable, covered by re-signing **CODE** a code literal — changing it needs a patch

EEPROM unit coding, not in the flash

CONVENTION	MEANING
0.01 km/h	Speeds/slip × 100. 5000 = 50 km/h.
0.01 g	Decel × 100, signed. -116 ≈ -1.16 g (unit unproven).
Q10	Value ÷ 1024 — lengths (m) and curve slopes.
~0.01 bar	Modelled pressure; consistent with evidence, not pinned (unproven).
variant 0–11	EEPROM field 0x4031AA+1 picks one column of the 12-wide model tables (7 distinct).
curve	Piecewise-linear (lo,hi,n,breaks,intercepts,slopes) read by 0x710FC : out = clamp(c + in·k/1024, lo, hi).

Architecture

Once per 10 ms frame the main task walks a fixed 41-call sequence: read sensors, estimate vehicle state, let each controller decide, arbitrate per wheel, run the pressure model, drive the valves, and send a torque request to the engine.



Editing & re-signing

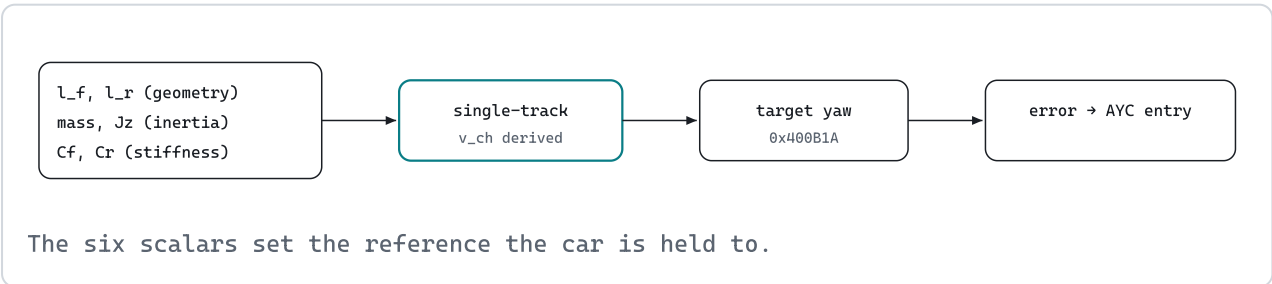
An edited image is rejected until two integrity values are recomputed — both solved and toolod.

LAYER	WHAT	FIX
ROM self-test	Hardware MISR over three regions; calibration region <code>[0x40300, 0x4249C]</code> , expected word at <code>0x4249C</code> .	Recompute after any cal edit.
BMV signature	512-bit RSA (e=7) over the application, in the trailing BMV block. Key recovered.	Re-sign after the MISR fix.

One command: `python tools/bmy_resign.py fix <edited.bin>` recomputes the MISR words then the BMV signature; `verify` reports both. Whether the bootloader enforces the signature at flash time is in the absent `0x0-0x47FFF` region — treat edited images as research-only.

Vehicle model

The yaw controller runs a single-track (bicycle) model. These scalars are its physical constants, stored as 12 columns — one per coding variant — loaded by `0x5DE44 / 0x5E424` . They set how much yaw the car should make for a given steering and speed; the controller acts on the error vs measured yaw.



PARAMETER	ADDR	STOCK	RAISE IT →	
l_f front axle→CG	0xD6F42	1413 Q10 m	CG treated further forward; shifts understeer balance.	CAL
l_r CG→rear axle	0xD6F5A	1413 Q10 m	l_f+l_r = wheelbase 2.760 m (fixed); trades balance with l_f.	CAL
mass	0xD6F72	1787 kg	Model assumes a heavier car: larger yaw-moment demand.	CAL
Jz yaw inertia	0xD6F8A	2999 kg·m ²	Model expects slower rotation → reacts earlier to yaw build-up.	CAL
Cf front stiffness	0xD6ECA	7527	More assumed front grip; lowers v_ch.	CAL
Cr rear stiffness	0xD6EE2	10596	More assumed rear grip; Cr>Cf keeps the model understeering.	CAL
v_ch characteristic speed	derived	105–142 km/h	Not stored — computed from the above. Higher = flatter yaw gain with speed = less yaw demanded at speed. M3 > 1M.	DERIVED
track front / rear	0xD6FA2/BA	1.538/1.536 m	Track geometry — Q10 metres (= real M3 track), not mm.	CAL
active variant	0x4031AA+1	EEPROM	Selects which column of all the above is used.	EEPROM

Wheel speed & tyres

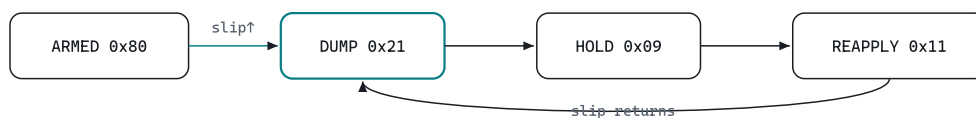
The four wheel speeds are the root of everything — vref, slip and the broadcast vehicle speed derive from them. Tyre circumference is the scale factor; the M3 ships the same geometry as the 1M.

PARAMETER	ADDR	STOCK	RAISE IT →	
circumference front	0x41CF4	2073 mm	ECU reads a higher road speed for the same wheel rpm — shifts vref and the speedo. The knob to match a different rolling radius.	CAL
circumference rear	0x41CF6	2073 mm	Same, rear. Front≠rear biases the undriven/driven comparison TCS uses.	CAL
tooth count F/R	0x41CF8/FA	48 / 48	Encoder teeth per rev; must match the tone ring.	CAL
standstill threshold	0x41CF2	72 (0.72 km/h)	Speed below which a wheel counts as stopped.	CAL

Sensor type: stock M3 and 1M images configure all four channels for VDA active sensors. There is no coding or calibration selector. **E46-type 2-level (7/14 mA) sensors work with a 4-byte firmware patch** (bench-verified on all four wheels, 2026-10-06): ASIC mode mask 0x28C → 0, both-edge timer capture, direction mask 0, presence mode 2. See [docs/E46_SENSOR_FIRMWARE_PATCH.md](#).

ABS slip & deceleration

ABS arms a wheel when its slip or deceleration crosses a speed- and grip-dependent threshold. The per-wheel phase state machine then cycles dump → hold → reapply.



Arming thresholds decide entry; the machine then cycles dump/hold/reapply. Pressure amounts are in the next section.

Key detail: the real arming threshold is the gross-slip curve at 0xD6CE2 (≈4 km/h low-speed, rising to ~11 % of vref) with a persistence counter. The small 0x40DB6 "entry-slip" curve (1.3–2.0 km/h) is used only inside the pair-logic partner test.

PARAMETER	ADDR	STOCK	RAISE IT →	
gross-slip arming curve	0xD6CE2	400 → ~11% vref	Slip a wheel must reach to arm ABS. Higher = later ABS, more bite on grip, worse low-μ.	ROM
entry-slip (pair test)	0x40DB6	≈1.3–2.0 km/h	Partner-in-slip test inside pair logic.	CAL
decel base	0x40620	–116 (≈–1.16 g)	Baseline decel threshold. More negative = tolerates harder decel first.	CAL
decel floor	0x4061E	–240 (≈–2.4 g)	Most-negative limit.	CAL
accel gain F/R	0x4061A/1C	40 / 25	How strongly wheel accel feeds the threshold builder.	CAL
low-speed floors	0x40644/5C	12 cols	Per-variant floors <20 / <60 km/h. Variants 10/11 permissive.	CAL

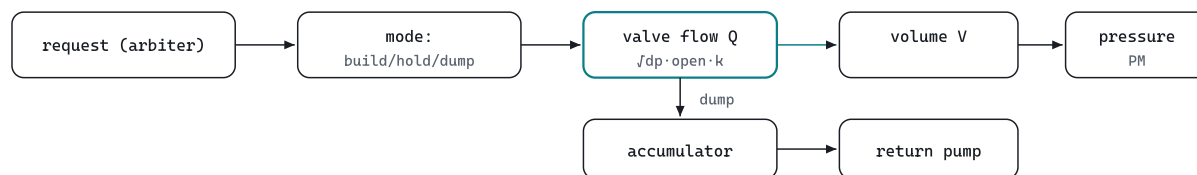
PARAMETER	ADDR	STOCK	RAISE IT →	
speed/g-term curves	0x40412/0x40674	12 variant copies	Shape threshold vs speed and vs g. 0–9 match the 1M; 10/11 differ.	CAL

Decision code 0x408DDE classifies each cycle: 1 no rear control · 16→8 select-low rear build · 2 pair-hold · 32 yaw-limited build (0–100 % lag) · 256 cornering reapply.

M3 vs 1M: core slip/decel targets are identical for variants 1–9; only 10/11 are more permissive at high speed. The M-car's licence lives in the yaw/DSC-mode logic, not these numbers.

Pressure & hydraulics

The controller works through a **volume model**: each wheel's fluid volume is tracked, converted to pressure through the COA p–V curves, and changed by valve flow $Q = \sqrt{\Delta p} \cdot \text{opening} \cdot k$. A low-pressure accumulator holds dumped fluid; a return pump drains it.



Pressure is a consequence of modelled volume, not a direct command.

PARAMETER	ADDR	STOCK	RAISE IT →	
apply ramp / cycle	0x403A8	400	Pressure built per 10 ms during reapply. Higher = faster, harsher re-clamp.	CAL
dump reductions	0x40E64	1000/1500/2000	Pressure released per dump stage. Higher = bigger release, quicker recovery, softer ABS.	CAL
reapply nudge	0x40D90	80	Small step entering reapply.	CAL
k_in build flow F/R	0x41B26	957 / 421	Inlet-valve flow gain. Higher = pressure builds faster for a given Δp.	CAL
k_out dump flow F/R	0x41B30	755 / 465	Outlet-valve flow gain. Higher = dumps faster.	CAL
LPA pressure curve	0x41B16	0, 130, 130, 500	Accumulator model (best clue unit ≈0.01 bar: 1.3–5.0 bar).	CAL
pump ramp / gain	0x41B34/36	1150 / 100	Return-pump delivery model.	CAL

PARAMETER	ADDR	STOCK	RAISE IT →	
pressure clamps	code	-8000 / 25000	Hard min/max on the request.	CODE
valve hold/boost current	code	1100/1524/1532	Solenoid currents for hold/boost/release.	CODE

Traction (TCS)

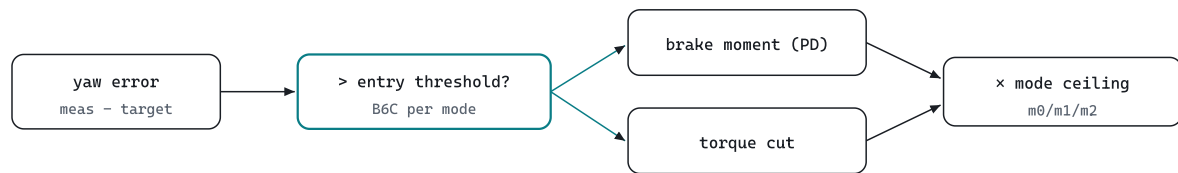
TCS acts on the driven (rear) wheels. Drive slip = rear – same-side front (0.01 km/h). Past the target curve, a PI loop requests engine-torque reduction and, if needed, brake pressure. The DSC mode (0/1/2) picks the set.

PARAMETER	ADDR	STOCK	RAISE IT →	
drive-slip target (set B)	0xF625C	curve/mode	Allowed rear-over-front slip before TCS cuts. Higher = more wheelspin permitted . Default set.	CAL
drive-slip target (set A)	0xF61F6	curve/mode	Alternate set for the surface/condition flag.	CAL
slip-target scalar	0x41046	2.3→10 km/h	Base slip target vs speed.	CAL
PI gains	0x41072	180/100/60/25	Torque-reduction loop gain. Higher = more aggressive cut.	CAL
PI P / I clamps	0x41068/6E	±2500 / ±200	Limits on the P and I terms.	CAL
gross-slip limit	code	min(1500, 4000-vref)	Hard ceiling — on the M3 a code literal (cal on the 1M).	CODE
DSC mode index	0x4031B1	0/1/2 live	Picks the target column; from DSC-button state + coding.	EEPROM

The cut leaves as CAN TX 0x0B6 to the DME (÷5 scale), merged with MSR + AYC by `dme_torque_request_compose` 0xCC2C2.

Yaw control & DSC mode

Where the M-car's character lives. AYC compares measured yaw to the model target; past the entry threshold it brakes individual wheels and cuts engine torque. Three DSC modes scale how much it allows — **mode 1 is by far the most permissive**.



Entry threshold and the mode ceilings are the primary "how loose is DSC" knobs.

PARAMETER	ADDR	STOCK	RAISE IT →	
entry threshold base A/B	0x413BA/BC	4.0 / 8.0 deg/s	Yaw error needed to trigger. Higher = more yaw/slip-angle allowed before DSC acts. (1396/2792 counts; LSB 0.0028648 deg/s)	CAL
torque ceiling per mode	0x4120A	m0 400 · m1 2000 · m2 330	Max engine-torque intervention. m1's 2000 is the permissive mode.	CAL
entry ceiling per mode	0x41424	30/30/23 deg/s	Clamp on the entry threshold. m2 tightest.	CAL
P-gain % per mode	0x41460	30→100 / 70→100	How hard it pushes once triggered.	CAL
speed factor per mode	0x41490/F0	curves ÷128	Scales the entry threshold with speed.	CAL
feature enable per mode	0xD6DA4	1 / 0 / 0	A feature active only in m0.	CAL
active DSC mode	0x4031B1	live	Chosen at runtime from DSC-button state + coding bit 0x4031F4 & 0x80.	EEPROM

Where "MDM" lives: the AYC tables are byte-identical to the 1M. The M-specific permissiveness is *which mode* the coding bit selects (m1 vs m2), not a different number. The mode ceilings are the lever, not the slip/decel targets.

How AYC brakes: the PD moment picks one wheel by the sign of the yaw error — **oversteer brakes the outer front; understeer the same-side rear** (rear-share 0x4155E, stock rear-only; front-outer pre-fill 0x411B4 = 5 bar). The engine-torque cap min-merges into the same TX 0x0B6 request as TCS. The yaw **observer** (0x5EBC4) is a nonlinear single-track model with a per-variant friction-limited tyre-force knee.

Brake coding

Brake hardware is described by pressure↔volume curves, selected by a 3-bit front and 3-bit rear coding field. Rotor and piston geometry are folded into these curves. The blocks are byte-identical to the 1M; coding picks code 0 vs 1 per axle.

PARAMETER	ADDR	STOCK	CHANGES	
p-V pressure axis	0x41B4E	0...32700	Shared breakpoints for the volume curves.	CAL
front volume set 0/1	0x41B62/76	10-pt curve	Front caliper fluid intake vs pressure — effective front brake "size". Code +3 selects.	CAL
rear volume set 0/1	0x41B8A/9E	10-pt curve	Rear equivalent; code +4.	CAL
BCO thresholds	0x41764...	pairs 0/1	Brake pressure-compare thresholds, per code.	CAL
brake coding F/R	0x4031AA+3/+4	EEPROM	Selects which curve set each axle uses.	EEPROM

Steering (CSI)

Steering angle (F-CAN 0x0C9) is converted to a road-wheel angle through a variable steering-ratio curve, then fed to the yaw reference. One of the few genuinely M3-specific calibrations — it grew and the ratio curve differs.

PARAMETER	ADDR	STOCK	RAISE IT →	
steer-ratio curve	0x40ECC	15.9:1 → 12.6:1	Road-wheel angle per steering angle vs lock. Re-scales the yaw-reference input.	CAL
ratio breakpoints	0x40EB4	0...10880 counts	Steering-angle axis (3 coding-selectable sets).	CAL
sensor sign tables	0x40EE4/F14	-1 (all)	Per-variant polarity for the yaw/lat-g channels.	CAL

DDS · LVC · faults · CAN inputs

PARAMETER	ADDR	STOCK	WHAT IT DOES	
DDS/RPA enable	0x41D48/4A	0 (use coding)	Indirect TPM: 40-bin wheel-speed resonance spectrum (14–151 Hz) watched for a shift vs a learned baseline. Non-zero forces on.	CAL
LVC hold count	0x4226A	75 (1M: 50)	The one M3-only scalar in LVC — a hold/delay count (~0.75 s).	CAL
CSW wheel reference	0x41E40	1562	Only live word in CSW — a DDS wheel reference.	CAL
direction fault ids	0xF576C	per wheel	DTCs for a wheel-direction fault. Lamp state is CAN-only (no GPIO lamp).	ROM

Inputs: DME torque on PT-CAN 0x0A8/0A9/0AA, gear on 0x0BA, steering on F-CAN 0x0C9, yaw/lat-g on F-CAN 0x0CD/0D1. The external torque-limit (0x78F) and several RX frames (0x0AC/0B4/0D5/1B4/388, F-CAN 0x118/11F) are present but disabled or dead in this calibration. The DSC also gateways some F-CAN frames onto PT-CAN.

Sources: FACTS.md · analysis/7846816A_symbols.md · xdf/MK60E5_7846816A.xdf. Values byte-verified against 7846816A unless marked med/low. Addresses are CPU addresses; file offset = CPU + 0x8000. Research only.